CADET _____ SECTION _____TIME OF DEPARTURE _____

DEPARTMENT OF CHEMICAL & BIOLOGICAL SCIENCE & ENGINEERING

CH365 2026-2027
Carnot Cycle Bonus
29 October 2026

TEXT: Smith, Van Ness, Abbott & Swihart
SCOPE: Lessons 22-23
TIME: 60 minutes

References Permitted: Open notes, book, internet, CHEMCAD, Mathematica, Excel.

INSTRUCTIONS

1. This is a BONUS exercise and is due **2359 29 October 2026**.
2. There are 2 problems on 1 page in this exercise (not including the cover page).
3. Save all electronic work in Canvas.

(TOTAL WEIGHT: 20 POINTS)

DO NOT WRITE IN THIS SPACE

PROBLEM	VALUE	BONUS
A	10	
B	10	
TOTAL BONUS	20	

Cadet:_____

Problem:	Weight:
A	10

A piston contains 0.1 kmol of nitrogen gas initially at 1.00 bar, 5.00 m^3, and 601.4 K. The gas undergoes a cyclic Carnot-type PV process between 601.4 K and 721.7 K, with minimum volume of 1.50 m^3 at pressure 4.00 bar. In other words, referring to Figure 5.2 on page 185, point d in the figure corresponds to 5.00 m^3 and 1.00 bar, and point b is 1.50 m^3 and 4.00 bar. Use Mathematica to solve for the intermediate points (points a and c in Figure 5.2) and construct a graph of the given Carnot cycle in Mathematica. Your plot should be professional in appearance and appear as shown in the sample plot below.

Problem:	Weight:
B	10

Use your results from Problem A to calculate the heat absorbed by the system from the hot reservoir, the heat ejected from the system to the cold reservoir, and the work produced, all in units of kJ. Calculate the efficiency of the Carnot cycle by both equations 5.6 and 5.7.

Additional information for Problems A and B:

Direction of cycle: a-b-c-d-a

$R=8.314$ J/(mol·K) = 0.08314 (bar·m^3)/(kmol·K)

$C_P=7R/2$, and

$C_V=5R/2$

Sample plot created in Mathematica:

